

Using a Single Sign-on in Java based Web Applications

Authentication to several related but independent software systems by logging in with a single user ID and password is called single sign-on. It is an access control property, whereby the user can seamlessly gain access to multiple connected systems without separately signing in with different user names and passwords.



Web application development is one of the most popular areas in IT software development. With clients becoming smarter, design architects and solutions providers are spending most of their time in the research and development of quicker, cheaper, more flexible and comprehensive solutions to suit any complex need of the user.

Of the many aspects of Web application development (or Web apps in a techie's language), the most challenging and interesting sub-section is the authorisation and sign-on facility for a Web app.

The most accepted and common way of authorisation is the log in/password, which the user has to supply. This is counter-matched with the records in the store (a database, usually) and permission is granted based on criteria like roles and permissions.

As customer demands grow more complex, the situation becomes challenging when the development

community is given more than one Web app interlinked for different purposes, and is asked to provide a sign-on for them.

This is when a single sign-on (SSO) makes sense. This article looks at its exact meaning, why it is required and how it got its current shape. We will also briefly discuss the various facilities/technologies available nowadays for the single sign-on.

SSO: In the early phase of adoption

To handle authentication for different Web apps using single sign-on information, certain features were proposed earlier. Two of these are listed below.

Bypass login: A bypass login is a crude workaround to handle logins for multiple related Web apps. It handles a login to the second application with a flag as a cookie set during the first Web app login. If this flag is set, then the second Web app will log in automatically. There is a huge problem in this method as the second Web app will not be role or